Functions of One-Dimensional Random Variables

Probability Theory and Mathematical Statistics

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Chapter 2, Section 4

Outline

1. Definition and Core Idea
2. Discrete Case
3. Continuous Case: CDF Method
4. Continuous Case: Formula Method
5. Piecewise Monotone Transformations
6. An Important Example: $Y = F(X)$
7. Summary and Exercises

Motivation and Definition

Why study functions of a random variable?

In applications, we often know the distribution of X but care about $Y = g(X)$. For example:

- Given the distribution of the radius X, find the distribution of the area $Y = \pi X^2$;
- Given the distribution of voltage X, find the distribution of power $Y = X^2/R$.

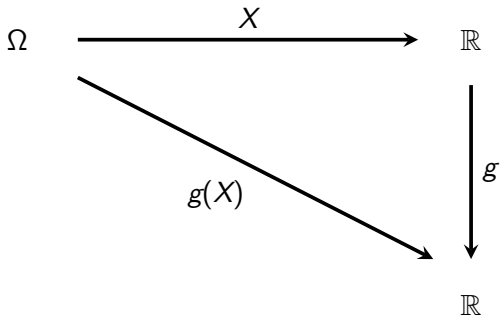
Definition 4.1

Let X be a random variable on Ω and $g : \mathbb{R} \to \mathbb{R}$ a real-valued function. Define

$$Y(\omega) = g\big(X(\omega)\big), \qquad \omega \in \Omega.$$

Since $X(\omega)$ is a real number and $g(X(\omega))$ is also a real number, Y is again a random variable on Ω. We call Y a **function of the random variable** X, denoted $Y = g(X)$.

Composite Mapping View



Key idea

The sample point ω is first mapped by X to a real number $X(\omega)$, then mapped by g to a real number $Y(\omega) = g(X(\omega))$. Thus $Y = g(X)$ is a composite function

$$\Omega \xrightarrow{X} \mathbb{R} \xrightarrow{g} \mathbb{R}.$$

Core Idea for Finding the Distribution of $Y = g(X)$

Convert probabilities about Y into probabilities about X

For any real y,

$$\{Y \leq y\} = \{\omega \in \Omega : g(X(\omega)) \leq y\} = X^{-1}\big(\{x \in \mathbb{R} : g(x) \leq y\}\big).$$

Therefore,

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

is obtained by converting the event $\{g(X) \leq y\}$ into an event about X, then using the distribution of X.

Two cases

- X discrete: $Y = g(X)$ is also discrete; compute its PMF directly;
- X continuous: $Y = g(X)$ is usually continuous; use the CDF method or the formula method.

Function of a Discrete Random Variable: Setup

Setting

Let X be a discrete random variable with probability mass function (PMF):

$$P(X = x_k) = p_k, \qquad k = 1, 2, \ldots$$

Consider a new random variable defined by a function $Y = g(X)$.

Possible Values of Y

Since X can only take discrete values $x_1, x_2, \ldots$, the new random variable Y is also discrete. Its possible values are the images of these points under g:

$$\{y_j = g(x_k) : k = 1, 2, \ldots\}.$$

Note that multiple x_k may map to the same y_j, so the set of possible values for Y may have fewer elements than the set for X.

Function of a Discrete Random Variable: PMF and Key Step

Computing the PMF of Y

To find the probability that Y takes a specific value y_j, we sum the probabilities of all x_k that map to y_j:

$$P(Y = y_j) = P(g(X) = y_j) = \sum_{k:\, g(x_k)=y_j} P(X = x_k) = \sum_{k:\, g(x_k)=y_j} p_k.$$

Crucial Reminder

If different values x_k map to the same value y_j, you must add their probabilities together.

This step is the most common source of errors. Always group all pre-images of y_j before summing.

Example 4.1 (Statement)

Example 4.1

Let

X	-1	0	1
P	$1/3$	$1/3$	$1/3$

Find the PMF of $Y = X^2$.

Example 4.1 (Solution)

Solution

Possible values of Y:

$$g(-1) = 1, \quad g(0) = 0, \quad g(1) = 1,$$

so $Y \in \{0, 1\}$.

$$P(Y = 0) = P(X = 0) = \frac{1}{3},$$

$$P(Y = 1) = P(X = -1) + P(X = 1) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

Hence

Y	0	1
P	1/3	2/3

Example 4.2 (Statement)

Example 4.2

Let

X	1	2	3	4	5	6
P	1/6	1/6	1/6	1/6	1/6	1/6

Find the PMF of $Y = (X-3)^2$.

Example 4.2 (Solution)

Solution

Compute Y for each X:

X	1	2	3	4	5	6
$Y=(X-3)^2$	4	1	0	1	4	9

So $Y \in \{0, 1, 4, 9\}$.

$$P(Y=0) = P(X=3) = \frac{1}{6},$$

$$P(Y=1) = P(X=2) + P(X=4) = \frac{1}{3},$$

$$P(Y=4) = P(X=1) + P(X=5) = \frac{1}{3},$$

$$P(Y=9) = P(X=6) = \frac{1}{6}.$$

Example 4.3 (Statement)

Example 4.3

Let $X \sim B(n, p)$ and $Y = n - X$. Find the PMF of Y.

Example 4.3 (Solution Part 1: Variable Substitution)

Setting up the PMF of $Y = n - X$

For any possible value $k = 0, 1, \ldots, n$, the probability mass function (PMF) of Y is defined as:

$$P(Y = k) = P(n - X = k).$$

Rearranging the equation inside the probability gives:

$$P(Y = k) = P(X = n - k).$$

Substituting the Binomial PMF

Since $X \sim B(n, p)$, we plug in its PMF at the value $n - k$:

$$P(X = n - k) = \binom{n}{n - k} p^{n-k}(1 - p)^k.$$

Example 4.3 (Solution Part 2: Simplification)

Using the symmetry of binomial coefficients

We apply the symmetry property $\binom{n}{n-k} = \binom{n}{k}$ to the expression:

$$P(Y = k) = \binom{n}{n-k} p^{n-k}(1-p)^k.$$

This simplifies to:

$$P(Y = k) = \binom{n}{k} (1-p)^k p^{n-k}.$$

Conclusion

The resulting expression matches the standard form of a Binomial PMF where the success probability is $1 - p$. Therefore:

$$Y = n - X \sim B(n, 1-p).$$

Example 4.3 (Solution Part 2: Interpretation)

Physical Interpretation

- The random variable X represents the number of **successes** in n trials, where each trial succeeds with probability p.
- The new random variable $Y = n - X$ represents the number of **failures** in the same n trials.
- Since each trial fails with probability $1 - p$, the number of failures naturally follows a Binomial distribution with parameters n and $1 - p$.

Conclusion

This intuitive argument perfectly matches the algebraic derivation:

$$Y = n - X \sim B(n, 1 - p).$$

CDF Method (General Approach) - Steps

The Three-Step Procedure

To find the distribution of $Y = g(X)$ using the CDF method:

1. **Write the CDF of Y:**
$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y).$$
2. **Convert to a set involving X:**
Define the region $I_y = \{x \in \mathbb{R} : g(x) \leq y\}$. Then,
$$F_Y(y) = P(X \in I_y) = \int_{I_y} f_X(x)\, dx.$$
3. **Differentiate to get the density:**
$$f_Y(y) = \frac{d}{dy} F_Y(y) = F_Y'(y).$$

CDF Method - Applicability and Significance

Applicability

This method works for **both monotone and non-monotone** functions $g(x)$. It does not require g to have an inverse.

Significance

- It is the **most general** approach for finding the distribution of a function of a random variable.
- Other formulas, such as the monotone transformation formula, can be derived directly from this method.
- When g is not one-to-one, we simply integrate the joint density over a more complex region I_y.

Example 4.4 (Statement)

Example 4.4

Let $X \sim U[-1, 1]$. Find the PDF of $Y = X^2$.

Example 4.4 (Solution Part 1: Setup and Boundary Cases)

Setup

Given $X \sim U(-1, 1)$, the probability density function of X is:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Since X takes values in $[-1, 1]$, it follows that $Y = X^2$ takes values in $[0, 1]$.

Boundary Cases for the CDF $F_Y(y)$

- **If** $y < 0$**:** The event $\{X^2 \leq y\}$ is impossible, so $F_Y(y) = 0$.
- **If** $y \geq 1$**:** The event $\{X^2 \leq y\}$ is certain, so $F_Y(y) = 1$.

Example 4.4 (Solution Part 2: The Core Calculation)

If $0 \leq y < 1$

The event $\{X^2 \leq y\}$ is equivalent to $\{-\sqrt{y} \leq X \leq \sqrt{y}\}$. Therefore, the CDF is the integral of the uniform density over this symmetric interval:

$$F_Y(y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2}\, dx.$$

Evaluating the integral

Since the integrand is a constant, the integral is simply the length of the interval multiplied by the height:

$$F_Y(y) = \frac{1}{2} \cdot (\sqrt{y} - (-\sqrt{y})) = \frac{1}{2} \cdot 2\sqrt{y} = \sqrt{y}.$$

Example 4.4 (Solution Part 3: Summary of the CDF)

Combining all cases

We have analyzed the three intervals for y. Now, we can piece them together to form the complete cumulative distribution function of Y:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \leq y < 1, \\ 1, & y \geq 1. \end{cases}$$

Next Step

Having obtained the CDF $F_Y(y)$, the final step is to differentiate it with respect to y to find the probability density function (PDF) $f_Y(y)$.

Example 4.4 (Solution Part 3: Summary of the CDF)

Recall the CDF

From the previous steps, the cumulative distribution function of Y is:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \leq y < 1, \\ 1, & y \geq 1. \end{cases}$$

Next Step

To find the probability density function, we need to differentiate $F_Y(y)$ with respect to y on each piece.

Example 4.4 (Solution Part 4: Deriving the PDF)

Differentiating the CDF

For the non-trivial interval $0 < y < 1$, differentiate $\sqrt{y}$ with respect to y:

$$\frac{d}{dy}(\sqrt{y}) = \frac{1}{2\sqrt{y}}.$$

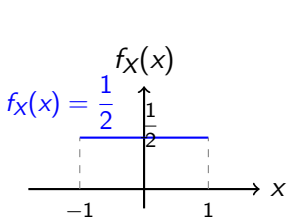
On the intervals $y < 0$ and $y \geq 1$, the derivative of a constant is 0.

Final Density Function

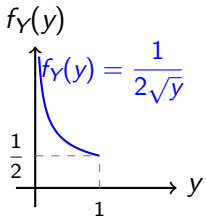
Combining the pieces, the complete probability density function of Y is:

$$f_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example 4.4 (Graph)



(a) $X \sim U[-1, 1]$



(b) $Y = X^2$

Interpretation

The uniform distribution on $[-1, 1]$ is transformed by $g(x) = x^2$ so that probability mass concentrates near $y = 0$. This is the effect of the two inverse branches $x = \pm\sqrt{y}$.

Example 4.5 (Statement)

Example 4.5

Let $X \sim N(0,1)$. Find the PDF of $Y = X^2$.

Example 4.5 (Setup and CDF)

Setup

The density of X is

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad -\infty < x < +\infty.$$

Since $X^2 \geq 0$, we have $Y \in [0, +\infty)$.

Step 1: Compute the CDF of Y

- If $y < 0$: $\{X^2 \leq y\} = \varnothing$, so
$$F_Y(y) = 0.$$
- If $y \geq 0$:
$$F_Y(y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \phi(x)\, dx.$$

Example 4.5 (Result and Conclusion)

Step 2: Differentiate to get the PDF

For $y > 0$, by the fundamental theorem of calculus (with the chain rule on the limits),

$$f_Y(y) = \frac{d}{dy} \int_{-\sqrt{y}}^{\sqrt{y}} \phi(x)\, dx = \frac{1}{2\sqrt{y}} \left[\phi(\sqrt{y}) + \phi(-\sqrt{y}) \right].$$

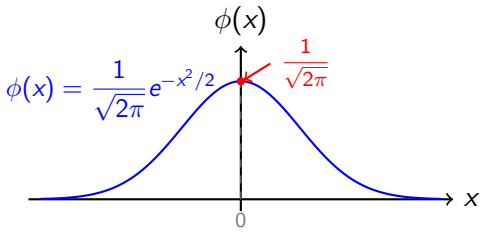
Since ϕ is even, $\phi(-\sqrt{y}) = \phi(\sqrt{y})$, so

$$f_Y(y) = \frac{1}{\sqrt{y}} \phi(\sqrt{y}) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}.$$

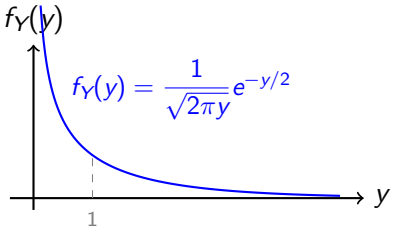
Result

$$f_Y(y) = \begin{cases} \dfrac{1}{\sqrt{2\pi y}} e^{-y/2}, & y > 0, \\ 0, & y \leq 0. \end{cases}$$

Example 4.5 (Graph)



(a) $X \sim N(0,1)$



(b) $Y = X^2 \sim \chi^2(1)$

Interpretation

The symmetric bell-shaped density is "folded" onto $[0, +\infty)$: the left and right halves of the probability mass are added together, so the density near $y = 0$ is very large.

Example 4.6 (Statement)

Example 4.6

Let $X \sim N(\mu, \sigma^2)$. Find the PDF of $Y = e^X$.

Example 4.6 (Solution Part 1: CDF Derivation)

Setup

Given $X \sim N(\mu, \sigma^2)$, its density is:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Consider the transformation $Y = e^X$. Since $e^X > 0$, the support of Y is $(0, +\infty)$.

Deriving the CDF

- **If $y \leq 0$:** The event $\{Y \leq y\}$ is impossible, so $F_Y(y) = 0$.
- **If $y > 0$:** Using the monotonicity of the exponential function and the natural logarithm, we have:

$$F_Y(y) = P(e^X \leq y) = P(X \leq \ln y) = \int_{-\infty}^{\ln y} f_X(x)\, dx.$$

Example 4.6 (Solution Part 2: Deriving the PDF)

Differentiating the CDF

For $y > 0$, differentiate $F_Y(y)$ with respect to y using the chain rule:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = f_X(\ln y) \cdot \frac{1}{y}.$$

Here, the factor $\frac{1}{y}$ comes from the derivative of the inner function $\ln y$.

Substituting the normal density

Plugging the normal density $f_X(x)$ at $x = \ln y$, we obtain the PDF of Y:

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma y} e^{-\frac{(\ln y - \mu)^2}{2\sigma^2}}, \qquad y > 0.$$

Example 4.6 (Solution Part 3: Log-Normal Distribution)

Definition: Log-Normal Distribution

If $X \sim N(\mu, \sigma^2)$, then the random variable $Y = e^X$ is said to follow a **Log-normal distribution**, denoted as:

$$Y = e^X \sim \mathrm{LN}(\mu, \sigma^2).$$

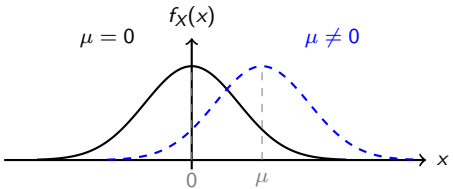
Applications

The log-normal distribution is widely used in:

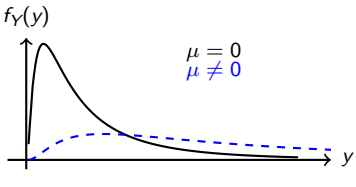
- **Economics:** modeling income distribution and stock prices.
- **Biology:** modeling growth processes and species abundance.
- **Reliability engineering:** modeling the lifetime of certain materials.

In general, it models **strictly positive** quantities that are the **product** of many independent, small positive factors (which corresponds to taking the log and obtaining a sum of many independent terms).

Example 4.6 (Graph)



(a) $X \sim N(\mu, \sigma^2)$



(b) $Y = e^X \sim \mathrm{LN}(\mu, \sigma^2)$

Interpretation

The exponential transform stretches the right tail: larger μ shifts the peak right and makes the right tail heavier.

Formula Method for Strictly Monotone g

Theorem 4.1 (Monotone Transformation)

Let X be continuous with PDF $f_X(x)$, and $P(a < X < b) = 1$ (with a, b possibly $\pm\infty$). Let $Y = g(X)$, where g is **strictly monotone and differentiable** on (a, b). Let (α, β) be the range of g, and let $g^{-1}(y)$ be its inverse. Then

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \dfrac{d}{dy} g^{-1}(y) \right|, & \alpha < y < \beta, \\ 0, & \text{otherwise.} \end{cases}$$

Formula Method for Strictly Monotone g

Proof idea

If g is strictly increasing, then

$$F_Y(y) = P(g(X) \le y) = P(X \le g^{-1}(y)) = \int_a^{g^{-1}(y)} f_X(x)\,dx.$$

Differentiating gives $f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$. The decreasing case adds a minus sign, which is absorbed by the absolute value.

Example 4.7 (Statement)

Example 4.7

Let $X \sim N(\mu, \sigma^2)$. Find the PDF of $Y = e^X$ using the formula method.

Example 4.7 (Setup)

Step 1: Check monotonicity

The function

$$g(x) = e^x$$

is strictly increasing and differentiable on $(-\infty, +\infty)$. So the monotone transformation formula applies.

Step 2: Inverse function and its derivative

The inverse of g is

$$g^{-1}(y) = \ln y, \qquad y > 0.$$

The range of g is

$$(\alpha, \beta) = (0, +\infty),$$

and the derivative of the inverse is

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{y}.$$

Example 4.7 (Computation Part 1)

Step 3: Apply the monotone transformation formula

For a strictly monotone transformation $Y = g(X)$, the density of Y is given by:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Here, $g(x) = e^x$, so the inverse function is $g^{-1}(y) = \ln y$. Its derivative is:

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{y}.$$

Substituting these into the formula yields:

$$f_Y(y) = f_X(\ln y) \cdot \frac{1}{y}.$$

Example 4.7 (Computation Part 2 and Result)

Step 4: Substitute the normal density

Recall that $X \sim N(\mu, \sigma^2)$, so its density is:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Final Density for the Log-Normal Distribution

Substitute $x = \ln y$ into the normal density and multiply by $\frac{1}{y}$. For $y > 0$, we obtain:

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma y} e^{-\frac{(\ln y-\mu)^2}{2\sigma^2}}.$$

This is the probability density function of the **Log-Normal distribution**.

Example 4.7 (Conclusion)

Conclusion

Therefore,

$$f_Y(y) = \begin{cases} \dfrac{1}{\sqrt{2\pi}\sigma y} e^{-\frac{(\ln y - \mu)^2}{2\sigma^2}}, & y > 0, \\ 0, & y \leq 0. \end{cases}$$

This agrees with the CDF method. The resulting distribution is the log-normal distribution, denoted $Y \sim \mathrm{LN}(\mu, \sigma^2)$.

Example 4.8 (Statement)

Example 4.8

Let X be continuous with PDF $f_X(x)$, and let

$$Y = aX + b, \qquad a \neq 0.$$

Find the PDF of Y.

Example 4.8 (Solution)

Solution

$g(x) = ax + b$ is strictly monotone, with inverse

$$g^{-1}(y) = \frac{y-b}{a}, \qquad \frac{d}{dy}g^{-1}(y) = \frac{1}{a}.$$

By the formula,

$$f_Y(y) = f_X\left(\frac{y-b}{a}\right) \cdot \left|\frac{1}{a}\right| = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

Special case: standardization of the normal

If $X \sim N(\mu, \sigma^2)$, take $a = 1/\sigma$, $b = -\mu/\sigma$. Then

$$Y = \frac{X-\mu}{\sigma} \sim N(0,1).$$

Linear Transformation Preserves Normality

Important fact

If $X \sim N(\mu, \sigma^2)$, then for any constants $a \neq 0$ and b,

$$Y = aX + b \sim N(a\mu + b,\ a^2\sigma^2).$$

Why this matters

- Linear transformations do not change the distribution type: both before and after are normal;
- The mean transforms as $a\mu + b$, the variance as $a^2\sigma^2$;
- When $a = 1/\sigma$, $b = -\mu/\sigma$, we obtain the standard normal $N(0, 1)$.

This is called the **linear transformation invariance** (or closure) of the normal distribution, and it is one of the main reasons the normal distribution is so widely used in statistical inference.

Piecewise Monotone Formula

Theorem 4.2

Let X be continuous with PDF $f_X(x)$ and $P(a < X < b) = 1$. Let $Y = g(X)$, where g on (a, b) can be decomposed into n disjoint subintervals $I_1, \ldots, I_n$, on each of which g is strictly monotone and differentiable. Let $g_k^{-1}(y)$ be the inverse of g on I_k. Then

$$f_Y(y) = \sum_{k=1}^{n} f_X\left(g_k^{-1}(y)\right) \left| \frac{d}{dy} g_k^{-1}(y) \right|,$$

with the convention that a term is 0 if y is not in the range of g on I_k.

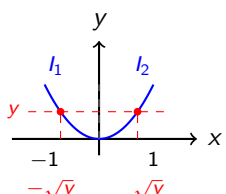
Idea

Add the contributions from all inverse branches. This is exactly the "folding" effect when a non-monotone function maps several x values to the same y.

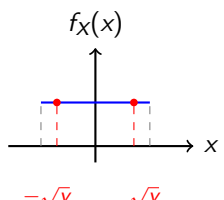
Example 4.9 (Statement)

Example 4.9

Let $X \sim U[-1, 1]$. Find the PDF of $Y = X^2$ using the piecewise monotone formula.



(a) $y = x^2$ on $[-1, 1]$



(b) Density and two inverse points

Example 4.9 (Setup)

Setup

$X \sim U[-1, 1]$, so its density is

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Since $X \in [-1, 1]$, we have $Y = X^2 \in [0, 1]$.

Idea

On $[-1, 1]$, the function $g(x) = x^2$ is not monotone, so we cannot apply the monotone formula directly. We first split $[-1, 1]$ into monotone pieces, then apply the piecewise monotone formula.

Example 4.9 (Step 1: Splitting into Monotone Pieces)

Decomposition of the domain

On the support $[-1, 1]$, the function $g(x) = x^2$ is not one-to-one. We must split it into two monotone pieces:

- **Left branch:** $I_1 = [-1, 0]$, where $g(x)$ is strictly decreasing. The inverse function is:

$$g_1^{-1}(y) = -\sqrt{y}.$$

- **Right branch:** $I_2 = [0, 1]$, where $g(x)$ is strictly increasing. The inverse function is:

$$g_2^{-1}(y) = \sqrt{y}.$$

Example 4.9 (Geometric Consequence)

Intersection with the parabola

For a fixed $y \in (0, 1)$, the horizontal line at height y intersects the parabola $Y = X^2$ at **two** points:

$$x_1 = -\sqrt{y}, \qquad x_2 = \sqrt{y}.$$

Contribution to the density

Because both x_1 and x_2 are pre-images of the same value y, the probability mass from both branches must be added together.

Therefore, the piecewise monotone formula for the density $f_Y(y)$ will be a sum of two terms, one from each branch.

Example 4.9 (Computation)

Step 2: Apply the piecewise monotone formula

For $0 < y < 1$, both the left and right branches contribute to the density:

$$f_Y(y) = f_X(-\sqrt{y}) \left| \frac{d}{dy}(-\sqrt{y}) \right| + f_X(\sqrt{y}) \left| \frac{d}{dy}(\sqrt{y}) \right|.$$

Step 3: Substitute f_X and the derivatives

On $[-1, 1]$, the density is constant $f_X(x) = \frac{1}{2}$. The absolute value of the derivatives is:

$$\left| \frac{d}{dy}(\pm\sqrt{y}) \right| = \frac{1}{2\sqrt{y}}.$$

Substituting these into the formula yields:

$$f_Y(y) = \frac{1}{2} \cdot \frac{1}{2\sqrt{y}} + \frac{1}{2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}}.$$

Example 4.9 (Remark on Symmetry)

Why are the two terms equal?

In the previous calculation, the two terms from the left and right branches are identical. This is due to two underlying symmetries:

- **Uniform Density:** The distribution of X is uniform on $[-1, 1]$, meaning $f_X(x) = \frac{1}{2}$ is a constant. Thus, $f_X(-\sqrt{y}) = f_X(\sqrt{y}) = \frac{1}{2}$.
- **Symmetric Inverse Branches:** The two inverse functions are $g_1^{-1}(y) = -\sqrt{y}$ and $g_2^{-1}(y) = \sqrt{y}$. Their derivatives have the same absolute value: $\left|\frac{d}{dy}(\pm\sqrt{y})\right| = \frac{1}{2\sqrt{y}}$.

Conclusion

Because both the density and the transformation are symmetric about $x = 0$, the two branches contribute equally to the final density of Y.

Example 4.9 (Result)

Probability Density Function of $Y = X^2$

Applying the transformation formula, we obtain the density of Y:

$$f_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example 4.9 (Consistency with the CDF Method)

Previously derived CDF

Using the CDF method earlier, we found:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \leq y < 1, \\ 1, & y \geq 1. \end{cases}$$

Verification

Differentiating $F_Y(y)$ with respect to y for $0 < y < 1$ directly yields:

$$f_Y(y) = F'_Y(y) = \frac{1}{2\sqrt{y}}.$$

Thus, the piecewise monotone transformation formula perfectly agrees with the CDF method.

Example 4.9 (Interpretation of the Result)

Why does the density blow up as $y \to 0^+$?

Observe that as $y \to 0^+$, we have $f_Y(y) = \frac{1}{2\sqrt{y}} \to +\infty$.

Intuitive Explanation

- The transformation is $g(x) = x^2$.
- Near $x = 0$, the derivative $g'(x) = 2x$ is very small.
- A small interval of x near 0 gets "squeezed" into an extremely tiny interval of y near 0.
- To maintain a total probability of 1, the density in this compressed region must become very large.

Therefore, the density blowing up at 0 reflects that the mapping $g(x) = x^2$ compresses probability mass toward $y = 0$.

Example 4.9 (Interpretation)

Interpretation

For a fixed $y \in (0,1)$, the horizontal line at height y intersects the parabola at two points:

$$x_1 = -\sqrt{y}, \qquad x_2 = \sqrt{y}.$$

Both points contribute to the density of Y, which explains why the piecewise monotone formula is a sum of two terms. The two contributions are equal here because f_X is constant on $[-1,1]$.

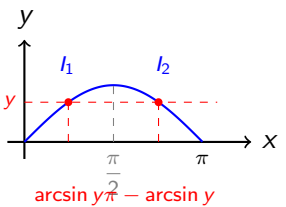
Example 4.10 (Statement)

Example 4.10

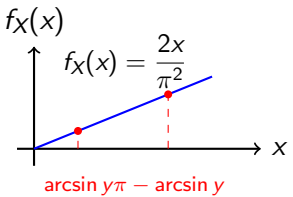
Let

$$f_X(x) = \begin{cases} \dfrac{2x}{\pi^2}, & 0 < x < \pi, \\ 0, & \text{otherwise.} \end{cases}$$

Find the PDF of $Y = \sin X$.



(a) $y = \sin x$ on $(0, \pi)$



(b) Density and two inverse points

Example 4.10 (Setup)

Idea

On $(0, \pi)$, the function $g(x) = \sin x$ is not monotone, so we cannot apply the monotone formula directly. We first split $(0, \pi)$ into monotone pieces, then apply the piecewise monotone formula.

Step 1: Split $g(x) = \sin x$ into monotone pieces

On $(0, \pi)$, the function $g(x) = \sin x$ splits into two monotone pieces:

- $I_1 = (0, \pi/2]$, strictly increasing, with inverse

$$g_1^{-1}(y) = \arcsin y;$$

- $I_2 = (\pi/2, \pi)$, strictly decreasing, with inverse

$$g_2^{-1}(y) = \pi - \arcsin y.$$

Example 4.10 (Derivatives)

Step 2: Range of Y and derivatives of the inverses

Since $X \in (0, \pi)$, we have $Y = \sin X \in (0, 1)$. For $0 < y < 1$,

$$\frac{d}{dy} g_1^{-1}(y) = \frac{1}{\sqrt{1 - y^2}}, \qquad \frac{d}{dy} g_2^{-1}(y) = -\frac{1}{\sqrt{1 - y^2}}.$$

Example 4.10 (Computation Part 1)

Step 3: Apply the piecewise monotone formula

For $0 < y < 1$, the random variable $Y = \sin X$ has two pre-images in the support of X: $x_1 = \arcsin y$ and $x_2 = \pi - \arcsin y$. Both pieces contribute to the density:

$$f_Y(y) = f_X(\arcsin y) \cdot \left| \frac{d}{dy} \arcsin y \right| + f_X(\pi - \arcsin y) \cdot \left| \frac{d}{dy}(\pi - \arcsin y) \right|.$$

Since the derivative of $\arcsin y$ is $\frac{1}{\sqrt{1-y^2}}$, this simplifies to:

$$f_Y(y) = \frac{f_X(\arcsin y) + f_X(\pi - \arcsin y)}{\sqrt{1 - y^2}}.$$

Example 4.10 (Step 4: Algebraic Simplification)

Substitute $f_X(x) = 2x/\pi^2$

Plugging the specific density of X into the formula, we get:

$$f_Y(y) = \frac{2\arcsin y}{\pi^2\sqrt{1-y^2}} + \frac{2(\pi - \arcsin y)}{\pi^2\sqrt{1-y^2}}.$$

Combine the numerators

Notice that both terms share the same denominator. Adding the numerators:

$$2\arcsin y + 2(\pi - \arcsin y) = 2\arcsin y + 2\pi - 2\arcsin y = 2\pi.$$

Therefore,

$$f_Y(y) = \frac{2\pi}{\pi^2\sqrt{1-y^2}} = \frac{2}{\pi\sqrt{1-y^2}}.$$

Example 4.10 (Final Result)

Recall the support of Y

Since X takes values in $(0, \pi)$, the function $Y = \sin X$ takes values in $(0, 1)$. Outside this interval, the density is zero.

Final Piecewise Density Function

Combining the density expression with its support, we obtain:

$$f_Y(y) = \begin{cases} \dfrac{2}{\pi\sqrt{1-y^2}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Distribution Function and Random Variable Composition

Key result

Let X be continuous with CDF $F(x)$. Then

$$Y = F(X) \sim U[0, 1].$$

This is the theoretical foundation of **inverse transform sampling**.

Interpretation

The CDF F "flattens" the distribution of X into a uniform distribution on $[0, 1]$. Conversely, if $U \sim U[0, 1]$, then

$$X = F^{-1}(U)$$

has CDF F. This is exactly the inverse transform method used in simulation.

Example 4.11 (Statement)

Example 4.11

Let X be continuous with CDF $F(x) = P(X \leq x)$. Define $Y = F(X)$. Prove $Y \sim U[0, 1]$.

Example 4.11 (Solution Part 1): Boundary Cases

Solution: Range of Y

Since $F(x)$ is a CDF, we know that $0 \leq F(x) \leq 1$. Therefore, the random variable $Y = F(X)$ must take values in $[0, 1]$.

Boundary Cases

- **If $y < 0$:** The event $\{Y \leq y\}$ is impossible. Thus,

$$F_Y(y) = P(F(X) \leq y) = 0.$$

- **If $y \geq 1$:** The event $\{Y \leq y\}$ is certain. Thus,

$$F_Y(y) = P(F(X) \leq y) = 1.$$

Example 4.11 (Solution Part 2): The Core Derivation

For the case $0 \leq y < 1$:

We define the **generalized inverse** of the CDF:

$$F^{-1}(y) = \inf\{x \in \mathbb{R} : F(x) \geq y\}.$$

Because $F(x)$ is non-decreasing, the event $F(X) \leq y$ occurs if and only if X is less than or equal to the threshold where the CDF first reaches y. That is,

$$\{F(X) \leq y\} \iff \{X \leq F^{-1}(y)\}.$$

Therefore, substituting this equivalence into the probability gives:

$$F_Y(y) = P(F(X) \leq y) = P(X \leq F^{-1}(y)) = F(F^{-1}(y)) = y.$$

Example 4.11 (Solution Part 3): Conclusion

Combining all cases

From the boundary cases and the core derivation, the CDF of $Y = F(X)$ is:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y, & 0 \leq y < 1, \\ 1, & y \geq 1. \end{cases}$$

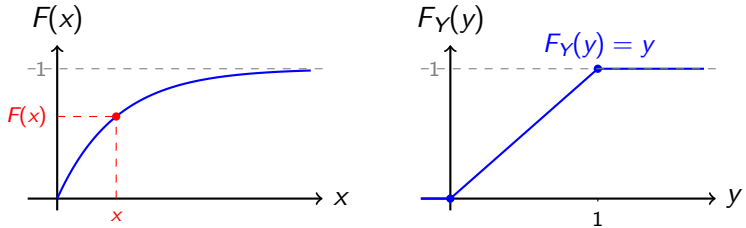
Final Conclusion

The resulting $F_Y(y)$ is exactly the cumulative distribution function of the uniform distribution on the interval $[0, 1]$.

Hence, we have proved that:

$$Y = F(X) \sim U[0, 1].$$

Example 4.11 (Graph)



(a) CDF F (b) $Y = F(X) \sim U[0, 1]$

Interpretation

No matter what distribution X originally follows, plugging it into its own CDF F produces a $U[0, 1]$ random variable. This is exactly why the inverse transform method works.

Summary (1): Function of a Random Variable

Definition

If X is a random variable and g is a real-valued function, then $Y = g(X)$ is also a random variable. It can be viewed as a composite mapping:

$$\Omega \xrightarrow{X} \mathbb{R} \xrightarrow{g} \mathbb{R}, \qquad \omega \mapsto X(\omega) \mapsto g(X(\omega)).$$

Core Idea

To find the distribution of $Y = g(X)$, we convert the probability of an event involving Y into the probability of an event involving X:

$$\{Y \leq y\} = \{g(X) \leq y\} = X^{-1}\big(\{x : g(x) \leq y\}\big).$$

This allows us to use the known distribution of X to compute probabilities for Y.

Summary (2): Discrete Case

Discrete Case

If X is a discrete random variable, then $Y = g(X)$ is also discrete.

- The possible values of Y are $y_j = g(x_k)$.
- The probability mass function (PMF) of Y is:

$$P(Y = y_j) = \sum_{k:\, g(x_k) = y_j} P(X = x_k).$$

Important Note

If different values of x_k map to the same value y_j, we must **add their probabilities** together.

Summary (2): Continuous Case and Common Results

Continuous case: three methods

- **CDF method** (general): first find

$$F_Y(y) = P(g(X) \leq y),$$

  then differentiate to get

$$f_Y(y) = F'_Y(y).$$

- **Formula method** (g strictly monotone):

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

- **Piecewise monotone**: split g into monotone pieces, apply the formula on each piece, and sum.

Summary (3): Continuous Case and Common Results

Common transformations

- Linear: $Y = aX + b$, then
$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$
- Standardization: if $X \sim N(\mu, \sigma^2)$, then
$$\frac{X-\mu}{\sigma} \sim N(0,1).$$
- Square: if $X \sim N(0,1)$, then
$$X^2 \sim \chi^2(1).$$
- Exponential: if $X \sim N(\mu, \sigma^2)$, then
$$e^X \sim \mathrm{LN}(\mu, \sigma^2).$$

Summary (4): Continuous Case and Common Results

Inverse transform sampling

If X is continuous with CDF F, then

$$Y = F(X) \sim U[0,1].$$

Conversely, if $U \sim U[0,1]$, then $X = F^{-1}(U)$ has CDF F.

Exercises (Part 1)

1. Let the PMF of X be

X	-2	-1	0	1	2
P	0.1	0.2	0.4	0.2	0.1

   Find the PMFs of $Y = 2X + 1$ and $Z = X^2$.
2. Let $X \sim B(3, 1/2)$. Find the PMF of $Y = 3 - X$, and identify the distribution of Y.
3. Let $X \sim U[0, 1]$. Find the PDF of $Y = -\ln X$, and identify the distribution of Y.
4. Let $X \sim N(0, 1)$. Find the PDF of $Y = 2X + 1$.

Exercises (Part 2)

5. Let the PDF of X be

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

   Find the PDF of $Y = X^2$.

6. Let $X \sim E(\lambda)$. Find the PDF of $Y = \sqrt{X}$.

7. Let the PDF of X be

$$f_X(x) = \frac{1}{\pi(1 + x^2)}, \qquad -\infty < x < +\infty \text{ (Cauchy)},$$

   Find the PDF of $Y = \arctan X$.

Thank you!